

AI Usage Note – Auto Categorization Webhook

Overview

Auto Categorization Webhook is an AI-powered ticket classification platform developed using Python, FastAPI, Pydantic, and Large Language Models (LLMs). The system automatically analyzes customer support tickets and predicts the appropriate category, subcategory, priority level, and confidence score. The objective of the project is to reduce manual ticket triaging effort, improve classification consistency, and accelerate support workflows.

What AI Helped With

Artificial Intelligence played a significant role in understanding and classifying support tickets. AI was used to:

- Analyze ticket titles and descriptions.
- Identify customer intent and issue type.
- Predict ticket categories such as Authentication, Billing, Technical Support, and Feature Requests.
- Generate subcategories for more precise classification.
- Assign priority levels (Low, Medium, High, Critical).
- Produce confidence scores to indicate prediction reliability.
- Generate structured JSON responses suitable for API integration.

By using AI, the system was able to classify tickets more accurately than traditional keyword-based approaches and significantly reduce manual effort.

What AI Got Wrong

Although AI was highly effective, several challenges were encountered during development:

Incorrect Categorization

Some ambiguous tickets were occasionally assigned to incorrect categories.

Example: A password-related issue was sometimes classified as general Technical Support instead of Authentication.

Inconsistent Output Formats

AI occasionally generated responses that did not match the expected JSON schema.

Low-Confidence Predictions

Short or unclear ticket descriptions sometimes resulted in uncertain classifications.

Over-Generalization

The model occasionally predicted broad categories rather than specific subcategories.

How These Issues Were Fixed

- Added few-shot training examples to improve context understanding.
- Implemented Pydantic schema validation for output verification.
- Introduced confidence threshold checks.
- Added retry mechanisms and error handling for API failures.
- Improved prompts to provide clearer classification instructions.

Best Prompt Used

The most effective prompt used in the project was:

Ticket Classification Prompt

"Analyze the following customer support ticket. Identify the most appropriate category, subcategory, and priority level. Return the result in structured JSON format along with a confidence score. Use the ticket title and description to understand the customer's intent and business impact."

This prompt consistently produced accurate and structured classification results while maintaining output consistency.

What I Learned

Through this project, I gained practical experience in:

- Prompt Engineering and AI Integration.
- FastAPI API Development.
- Pydantic Validation and Configuration Management.
- OpenAI API Integration.
- Error Handling and Retry Mechanisms.
- REST API Design and Testing.
- Automated Testing using Pytest.
- Type Checking using MyPy.
- Code Quality Validation using Ruff.
- Environment Variable Management using .env files.

Conclusion

The Auto Categorization Webhook project demonstrated how Artificial Intelligence can automate support ticket classification and prioritization. AI significantly improved efficiency,

reduced manual effort, and enhanced consistency in customer support workflows. At the same time, the project highlighted the importance of prompt engineering, validation mechanisms, testing, and human oversight to ensure reliable AI-driven solutions.